

Barriers to the development of the supply side of the organic food market in Poland

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Abstract. The chapter's research objective is to identify barriers to the development of the supply side of Poland's organic food market. It presents the results of the empirical survey conducted with two groups of respondents: organic food distributors ($N = 120$) and owners of organic farms ($N = 120$). We used statistical description, non-parametric Wilcoxon signed-rank test and exploratory factor analysis. The in-depth analysis of barriers included exploratory factor analysis, which allowed us to identify 4 groups of common factors for distributors and 4 for farmers. The results of the analyses for both groups of surveyed actors led us to the following conclusion: the most important barriers are bureaucratic and administrative difficulties, as well as a lack of adequate government support. Therefore, there is a need to simplify and stabilise the European Union subsidy system in Poland. Another important barriers arise from the specific character of agricultural production (the high costs and low profitability of organic production). They could be diminished thanks to the increase in the rates of organic farming subsidies. Market-led changes should involve better connections between farmers and distributors.

Keywords: organic farming, supply side of an organic food market, distribution in the organic food market

1 Introduction

In the years 2004-2026, the number of Polish organic farms increased from 3705 (Kociszewski & Graczyk, 2021) to 24209 (IJHARS, 2026a) – by 553% – and the organic utilised agricultural area (UAA) from 82.7 thousand ha to 757.095 thousand ha (IJHARS, 2026b) – by 815%. It means that Poland's accession to the EU influenced the dynamic development of organic production potential, reaching 5.3% of UAA. However, it remains smaller than in the European Union (EU-27) as a whole, where the share of organic crops in total UAA was 11.1% in 2024 (Willer et al., 2026). According to Żakowska-Biemans (2022, p. 22), the Polish organic food market would need to grow by 9.4% annually to reach around EUR 600 million by 2026. In 2024, it was about EUR 490 million, or 1% of the total value of the Polish food market (Biokurier, 2025). The questions which arise are: why the market value did not grow as it was predicted; why the organic market share in the total food market is still much smaller than in the entire EU, 4.5% (Willer et al., 2026); why production potential is smaller than in the Western European member states and in countries neighbouring Poland. For instance, in 2024, the share of organic crops in total UAA was 27.2% in Austria, 9.4% in France, 11.5% in Germany, 22.4% in Estonia, 17.2% in Czechia and 14.1% in Slovakia (Willer et al., 2026). To answer these questions, the analysis focused on the factors affecting organic food distributors and farmers in light of market development prospects. The research objective was to identify the barriers to the development of the supply side of the organic food market in Poland. The literature review identified the research gap and helped to formulate a hypothesis. The next parts of the chapter covered the research methods and the results, which were discussed for comparison with the studies included in the literature review.

2 Literature review

Barriers to the development of organic farming address a range of issues – from economic costs and operational complexity to regulatory uncertainty (Fadini et al., 2026). Based on the literature review, we distinguished production, economic/market, organisational and institutional, social, and informational barriers to the development of organic farming.

Production barriers include insufficient domestic production, irregular supply, and dependence on imports of processed products (Smoluk-Sikorska, 2021; Wägeli & Hamm, 2016; Götze et al., 2016; Scalvedi & Saba, 2018). Due to restrictions on the use of artificial fertilisers and pesticides, there are problems with an unstable raw material base, limited access to organic fertilisers, and their high prices, which entail the need to develop a model of cooperation with suppliers (Żakowska-Biemans et al., 2020, p. 276). Another problem is insufficient processing capacity (Dornberg et al., 2016).

Economic barrier relates to the producer-consumer price ratio. The prices of organic food are too high for consumers (Grzybowska-Brzezińska & Gorłowa, 2019, pp. 41–42) but, on the other hand, too low for the farmers (Koreleska, 2009). It results from high margins charged by intermediaries. Organic products are often more expensive than their conventional counterparts due to the relatively higher costs of organic production and distribution (Sadler et al., 2025). Organic farming relies on labour-intensive techniques, which increases operating costs (Kołoszko-Chomentowska & Stalgiene, 2019). The problem arises, among other things, from the small volume of organic products (Michalke et al., 2023). Prices of conventional products are undervalued due to the lack of environmental externalities (Michalke et al., 2023).

Organisational, administrative and institutional barriers include weak political commitment of public institutions (the state), the lack of strong organisations representing the political and economic interests of organic farming operators, and excessive bureaucratic burdens (Sołtysiak, 2021), including those related to organic certification, as well as complicated procedures and their costs (Górska-Walczak et al., 2025). In the USA, regulatory issues were found to be the most common reason for organic farmers switching back to conventional farming (Sierra et al., 2008). Polish farmers also indicate institutional and legal barriers: unexpected changes in organic farming regulations, decreasing level of public support for farm development and the formation of producer groups (Komorowska, 2014; Brodzińska, 2014; Nachtman, 2015; Gil, 2016; Łuczka & Kalinowski, 2020; Łuczka, 2024). In this respect, there are many administrative and bureaucratic barriers related to how the policy is implemented by national organisations that pay subsidies to organic farms (Najwyższa, 2019). The lack of government assistance and insufficient infrastructure also pose significant challenges to the effective implementation of sustainable production methods (Alhafeza, 2026).

Market, social and information barriers include the weak cooperation in the supply chain (Vlahović et al., 2019; Koreleska, 2017). The governmental Framework Action Plan for Organic Food and Farming in Poland for 2021–2027 (Ministerstwo, 2022) identifies barriers such as underdeveloped distribution channels and weak cooperation among actors responsible for the supply of organic food (European Commission, 2021; Kociszewski et al., 2024). One of the weakest links is organic food processing, which incurs high transaction costs (Górska-Walczak et al., 2025). In Poland, the main distribution channel is based on sales to small retail outlets. Wholesale trade in organic products remains underdeveloped, hindering the effective flow of products from processors to retailers.

The social context is crucial for the development of organic production – the knowledge and skills of farmers, processors, and agricultural advisors; openness to cooperation; consumer awareness; and a transparent flow of information between practitioners and institutions of agricultural policy (Kociszewski et al., 2024; NIK, 2019). In that context, there is a problem with the limited transfer of knowledge to agricultural practice, including a shortage of advisors specialising in organic farming (Sołtysiak, 2021).

Most studies on barriers to the development of organic farming focus on the demand side (Ghali, 2026; Castro Campos & Qi, 2024). Research on the food supply chain and its participants focuses on management. The literature review indicated a

research gap regarding the characteristics of supply-side barriers as perceived by distributors. Furthermore, there is a lack of detailed, statistically based elaborations on the hierarchy and specific strengths of barriers to the development of Polish organic production, based on farmers' opinions. Consequently, no papers compare the results gathered in these groups of actors. Considering the information presented in the literature review, we formulated the research hypothesis: bureaucratic and administrative difficulties and a lack of adequate public support are the biggest barriers to the development of the supply side of the Polish organic food market. To verify the hypothesis, we indicated the goal of the paper presented in the introduction.

3 The research methods

The article is based on the results of the empirical survey, which was projected and constructed by the authors. We prepared the questionnaires, and the study was carried out by a specialised company using the Computer Assisted Telephone Interview (CATI) method. The surveys were conducted in 2021 on a random nationwide sample of 120 respondents from a group of farmers certified for organic production ($N = 120$) and 120 distributors of organic food ($N = 120$). The number of interviewed farmers reflected the area structure of Polish organic farms and their distribution by Polish regions (voivodships). We used these results in our calculations conducted with IBM SPSS Statistics 27.0 or 28.0.

The research hypothesis was verified using descriptive statistics, the Wilcoxon signed-rank test, and exploratory factor analysis of responses from distributors and farmers. The respondents in the first group identified the main barriers to the development of distribution networks of Polish organic farming products and increased sales. Organic farmers identified the main barriers to production development and increased sales in Polish organic farming. The results obtained in both groups were analysed separately and compared.

The respondents assessed the barriers in terms of their significance on a 7-point scale (1 – very low significance; 7 – very high significance). To check which barriers played a significant role in the development of organic food distribution networks and organic production, we used the non-parametric Wilcoxon one-sample signed-rank test. It enabled us to check whether the location parameter (median in this study) of the responses equals a given value. We assumed that factors with a median exceeding 4 (a neutral value) were rated highly (strongly hindering the development of the organic market). The null hypothesis was that the median is less than or equal to 4, and it was verified against the alternative hypothesis - the median is greater than 4. For a large sample size ($n > 25$), the distribution of the T-statistic is approximately normal (Aczel, 2018, pp. 904–907). The standardised Z statistic of the study is as follows:

$$Z = (T - E(T)) / \sqrt{\text{var}(T)} \quad (1)$$

where $T = \min(S^+, S^-)$ – test statistic; S^+ – sum of ranks of positive differences, S^- – sum of ranks of negative differences; $E(T) = n(n + 1)/4$ – expected value; $\text{var}(T) = n(n + 1)(2n + 1)/24$ – variance¹.

We used exploratory factor analysis to identify the latent factor structure of barriers to the development of organic production. This analysis aims to find a new set of variables (fewer than the original set) that expresses the relationships among the observed variables (Kim & Mueller, 1978).

Table 1: The statistics of the Kaiser–Meyer–Olkin test, Bartlett's sphericity test, and reliability test – barriers to the development

The respondents	Number of factors	KMO adequacy measurement	Asymptotic chi-square statistic	df	<i>p</i> -value	Cronbach's Alpha	% of variance
Distributors	9	0.753	332.197	36	<0.001	0.808	73.4
Farmers	10	0.828	442.085	45	<0.001	0.852	74.2

Each observable variable is a linear function of unobservable variables (common factors) and a single specific variable. The coefficients of the linear combinations are the factor loadings. They determine how strongly the real variable correlates with the factors derived from the loadings. The Hotelling principal components method is used to estimate factor loadings iteratively. To facilitate the interpretation of the extracted principal factors, we performed an orthogonal Varimax rotation, which minimises the number of variables needed to explain a given common factor (Bednarska & Cypryańska, 2013).

The factor analysis was conducted separately on the responses from distributors and farmers. The validity of the use of the factor analysis is each time evidenced by the following: (i) the Spearman's rank correlation coefficients with statistically significant values indicating at least moderate correlations between pairs of factors; (ii) a high value of the Kaiser-Meyer-Olsen test (with KMO exceeding 0.75), proving that the correlation matrix is adequate to the assumptions of the factor analysis; and (iii) a statistically significant result of the Bartlett's sphericity test ($p < 0.001$), confirming that the correlation matrix as a whole contains significant correlation coefficients. The exploratory factor analysis was performed using principal component analysis with Varimax rotation (Kaiser normalisation). The scree plot criterion was used to determine the number of components to explain at least 70% of the common variance. Four components were usually extracted. The extracted common factors together explain 73.4% and 74.2% of the variance, which are quite high (Table 1).

The dimension reduction procedure included variables for which the Wilcoxon signed-rank test confirmed the significance of the barriers, and the mean significance score exceeded 4. The reliability and high consistency of the scale used, with

¹ Previously, we conducted a study on the opportunities for the development of the supply side of the organic market (Kociszewski et al., 2026). We used the same research methods.

the number of variables defined in this way, are evidenced each time by a high value of the Cronbach's alpha statistic (higher than 0.8) (Table 1).

4 Results

The factors perceived by distributors as barriers to the development of the Polish organic market were identified based on decreasing values of the average measures, i.e., the arithmetic mean, median, and mode (Table 2). The highest values of the median and the mode (5 and 7, respectively) related to five factors, i.e., high costs of organic production, bureaucratic and administrative difficulties, the lack of adequate state support, insufficient environmental awareness of consumers, and high prices of organic food discouraging consumers. The next barrier was "customers' preference for conventionally produced food", with a median of 5 and a mode of 5. The arithmetic mean values exceeded the neutral value (4) for 9 of the 15 barriers, indicating they were given high significance. The Wilcoxon signed-rank test indicated that 8 factors were assessed as significant barriers.

Table 2 Main barriers to the development of distribution networks of Polish organic farming products and increased sales – descriptive parameters and Wilcoxon signed-rank test, distributors' responses, $N = 120$

Specification	Mean	Median	Mode	Statistics Z	p -value
High costs of organic production	5.02	5	7	-5.315	<0.001
Bureaucratic and administrative obstacles for organic farms	5.00	5	7	-5.135	<0.001
Lack of adequate support for organic farming from the government	4.90	5	7	-4.479	<0.001
Too little environmental awareness among consumers	4.78	5	7	-4.225	<0.001
The difficulty of using organic farming methods	4.63	5	4	-3.729	<0.001
Customers' preference for conventionally produced food	4.48	5	5	-2.878	0.002
Too high organic food prices discouraging consumers	4.45	5	7	-2.236	0.013
Low profitability of production	4.48	4	4	-2.970	0.001
Low sales opportunities for organic products (too low demand)	4.28	4	5	-1.559	0.059
Low level of innovation of Polish agri-food companies	3.89	4	3	-0.545 ^a	0.293
Instability or uncertainty of the supply of agricultural products by Polish farmers	3.88	4	4	-0.838 ^a	0.201
Too little number of organic food processing plants	3.86	4	4	-0.866 ^a	0.193

Too little information about the offer from Polish farmers	3.78	4	2	-1.270 ^a	0.102
Too small amount of agricultural products offered by Polish organic farms	3.72	4	4	-1.759 ^a	0.039
Limited opportunities to contact organic farmers	3.30	3	1	-3.907 ^a	<0.001

^a Based on the sum of positive ranks.

Based on these factors, the test statistic was obtained from the sum of negative ranks; *p*-values for these factors are less than 0.05 (Table 2), indicating that distributors believe these factors are the main barriers to the development of distribution networks and to increased sales.

For six factors, the Z-test statistic was calculated using the sum of positive ranks. Note that the test statistic was statistically significant at the 0.05 level for two factors, i.e., insufficient produce on offer from organic farms and limited opportunities to establish contacts with farmers (Table 2). This means that distributors were more likely to give these factors a significance of less than the neutral value (4). The differences in mean positive and negative ranks are significant. Therefore, the distributors do not perceive these factors as barriers to the development of distribution networks.

The in-depth analysis of barriers includes an exploratory factor analysis that identifies the latent structure of the common factors. The dimensionality reduction procedure included 9 variables, of which 8 had barrier significance, as confirmed by the signed-rank test. One variable, despite the lack of statistical confirmation, had a mean significance rating greater than the neutral value (4).

Table 3 Matrix of rotated components¹ obtained using the principal component analysis with Varimax rotation with Kaiser normalisation – barriers, distributors' responses, *N* = 120

Barriers to the development of production and increased sales in Polish organic farming	Component			
	1	2	3	4
Bureaucratic and administrative obstacles for organic farms	0.878	0.196	0.029	0.114
Lack of adequate support for organic farming from the government	0.731	0.108	0.338	0.205
The difficulty of using organic farming methods	0.138	0.871	0.114	0.052
High costs of organic production	0.181	0.795	0.349	0.185
Low sales opportunities for organic products (too low demand)	0.059	0.272	0.844	0.179
Low profitability of production	0.443	0.271	0.682	0.034
Too high organic food prices discouraging consumers	0.007	0.237	-0.119	0.867
Too little environmental awareness of consumers	0.203	0.076	0.314	0.668
Customers' preference for conventionally produced food	0.295	-0.129	0.350	0.517

Notes: ¹ The rotation reached convergence in 5 iterations.

We identified 4 common factors corresponding to barriers according to distributors. (Table 3). The percentages of variance of the four components were similar: 18.7%, 18.5%, 18.5%, and 17.7%. The first common factor (legal and organisational barriers) comprised the following variables with high factor loadings: bureaucratic and administrative obstacles and inadequate government support. The second main factor included the variables with high factor loadings: difficulties in applying organic farming methods and high production costs. This component identified the barrier resulting from the specific character of organic farming. The third common factor (demand and profitability) comprised two variables. One of them (too low demand) has a strong correlation with the component. The fourth component loading three factors (including one with a high factor loading), reflects traditional consumer attitudes.

The results of the exploratory factor analysis converged with the hierarchy of barriers. The order of the common factors corresponds to the high ratings of factor significance given by the distributors. This leads to three groups of significant barriers to the development of distribution:

- The first group comprises external factors that determine the functioning of organic farms. It includes legal and administrative barriers comprising bureaucratic and administrative difficulties and the lack of adequate state support. The distributors most often gave these factors the highest rating (7). "Bureaucratic and administrative obstacles" is more strongly correlated with the common factor than "the lack of adequate support from the government".
- The second group comprises internal factors that determine the functioning of organic farms. It includes the barriers corresponding to the second and third common factors, i.e., barriers resulting from the specific character of organic production and the "demand and profitability" factor. Distributors most often gave these factors at least an average rating (4). "Difficulties in applying organic farming methods" was more strongly correlated with the "specific character of organic production" factor, while "too low demand" was more strongly correlated with the demand and profitability.
- The third group of factors describes traditional consumer attitudes (the fourth common factor). It includes the following barriers: low consumer environmental awareness, high prices that discourage consumers, and customers' preference for conventionally produced food. The first two factors were most often given the highest rating (7), while the third one was most often given a high rating (5). The factor "high prices" was more strongly correlated with the common factor than the other two factors.

In indicating the barriers to the development of production and increased sales in Polish organic farming, farmers used a 7-point scale. For almost all factors the median was at least 4. The median was below the neutral value (4) for only one factor (insufficient consumer demand for organic food). The factors perceived as barriers to production development and increased sales were identified based on decreasing values of the average measures, i.e., the arithmetic mean, median, and mode (Table 4). The highest values of the median and the mode (6 and 7, respectively) relate to five factors, i.e., bureaucratic and administrative difficulties, the lack of adequate

state support, excessively high costs of organic production, excessively low prices of produce offered to organic farms by processors, and difficulties in applying organic farming methods on farms.

Table 4 Main barriers to the development of production and increased sales in Polish organic farming – descriptive parameters and Wilcoxon rank-signed test, organic farmers' responses, $N = 120$

Specification	Mean	Median	Mode	Statistics Z	p -value
Bureaucratic and administrative obstacles for organic farms	5.33	6	7	-5.848	<0.001
Lack of adequate support for organic farming from the government	5.31	6	7	-5.742	<0.001
High costs of organic production	5.29	6	7	-6.106	<0.001
Too low prices of agricultural products offered to organic farms by processors, wholesalers or shops	5.23	6	7	-5.339	<0.001
Difficulties with weed control methods used in organic farming	4.97	6	7	-4.518	<0.001
Low profitability of production	5.01	5	7	-5.055	<0.001
Too little environmental awareness of consumers	4.82	5	7	-4.200	<0.001
Low yields	4.77	5	4	-4.590	<0.001
Too little number of organic food processing plants	4.43	5	7	-1.996	0.023
Customers' preference for conventionally produced food	4.41	4	4	-2.081	0.019
Too high organic food prices discouraging consumers	4.27	4.5	7	-1.170	0.121
Limited access to fertilisers and pesticides permitted in organic farming	4.10	4	4	-0.356	0.361
Limited opportunities to contact shops, wholesalers or processors	3.96	4	7	-0.002	0.499
Low level of innovation of Polish agri-food companies	3.89	4	1	-0.634 ^a	0.263
Too low demand for organic products from shops, wholesalers or processors	3.78	4	1	-1.026 ^a	0.152
Low sales opportunities for organic products (too low consumer demand)	3.60	3	1	-1.999 ^a	0.023

^a Based on the sum of positive ranks.

The next three factors are as follows (median: 5; mode: 7): low profitability of organic farming, insufficient consumer environmental awareness, and an insufficient number of organic food processors.

The Wilcoxon signed-rank test indicates that organic farmers believe there are 10 significant barriers to production development and increased sales. For these factors, the test statistic was obtained based on the sum of negative ranks; p -values for these factors are less than 0.05 (Table 4), indicating that farmers believe these factors are the main barriers to the development of production and increased sales. For the three factors, the Z-test statistic was calculated using the sum of positive ranks. The test statistic was statistically significant at the 0.05 level for only one factor (insufficient consumer demand for organic food). This means that farmers were more likely to give this factor a significance of less than the neutral value. The differences in mean positive and negative ranks are significant. Therefore, producers (farmers) do not perceive low demand as a barrier to production development and increased sales in Polish organic farming.

The exploratory factor analysis identified four components of barriers to the development of production and increased sales in Polish organic farming from the perspective of organic farmers (Table 5). Ten variables (barriers) were used. The Wilcoxon signed-rank test confirmed their significance. Three common factors have a similar percentage of variance explanation (21.6%, 19.4%, and 18.1%), while the fourth component explains 15.2% of the common variance.

Table 5 Matrix of rotated components¹ obtained using the principal component analysis with Varimax rotation with Kaiser normalisation – barriers, organic farmers' responses, $N = 120$

Barriers to the development of production and increased sales in Polish organic farming	Component			
	1	2	3	4
Low yields	0.800	0.268	-0.081	0.244
Difficulties with weed control methods used in organic farming	0.781	-0.093	0.277	0.200
High costs of organic production	0.613	0.385	0.471	-0.209
Low profitability of production	0.589	0.489	0.367	-0.046
Bureaucratic and administrative obstacles for organic farms	0.195	0.846	0.099	0.127
Lack of adequate support for organic farming from the government	0.054	0.776	0.309	0.235
Too little number of organic food processing plants	0.058	0.191	0.783	0.289
Too low prices of agricultural products offered to organic farms by processors, wholesalers or shops	0.283	0.229	0.705	0.197
Customers' preference for conventionally produced food	0.059	0.251	0.098	0.799
Too little environmental awareness of consumers	0.237	0.017	0.380	0.737

Notes: ¹ The rotation reached convergence in 5 iterations.

The first common factor comprises four variables, i.e., low yields, difficulties in applying organic farming methods, high costs of organic production, and low

profitability of farming. This component identifies the barrier resulting from the specific character of organic farming. The second common factor is loaded by two variables (bureaucratic and administrative difficulties and the lack of state support). It refers to legal and organisational barriers. The third component contains two variables, i.e., an insufficient number of processors and excessively low prices of agricultural produce. It defines the barriers of organic food processing. The last component defining barriers (conventional consumer attitudes) comprises two variables: a preference for conventionally produced food and low environmental awareness among consumers.

The results of the exploratory factor analysis largely converged, with the hierarchy of barriers. The order of the common factors usually corresponded to the high ratings of factor significance. This leads to three groups of significant barriers (the order of their appearance is determined by the correlations between the ratings of the individual factors revealed in the common factors):

- The first group comprises internal factors that determine the functioning of organic farms. They are identified as the first and third common factors. Farmers most often rated these factors very high. Only "low yields" were most often rated with the neutral value (4). That barrier is more strongly correlated with the "specific character of agricultural production", while "insufficient number of organic food processors" is more strongly correlated with "organic food processing".
- The second group comprises external factors that determine the functioning of organic farms. It includes legal and administrative barriers (the second common factor). The respondents most often gave these factors the highest rating (7). "Bureaucratic and administrative difficulties" is more strongly correlated with the common factor than "the lack of adequate government support".
- The third group of factors describes conventional consumer attitudes (the fourth common factor). "Little environmental awareness" was most often rated highest (7), while "preference for conventional food" was most often rated neutral (4). Despite the lower ratings, the last barrier is more strongly correlated with the common factor than "little environmental awareness".

5 Discussion

The hypothesis states that bureaucratic and administrative difficulties and the lack of adequate public support are the biggest barriers to the development of the supply of the Polish organic food market. Based on the results of the distributors' survey, the five key barriers with the highest values of the median and the mode (5 and 7) (Table 2) are as follows: high costs of organic production, bureaucratic and administrative difficulties, the lack of adequate government support, insufficient environmental awareness of consumers, and high prices of organic food discouraging consumers. The results of the in-depth factor analysis (Table 3) showed that distributors believed the key barriers to sales development were contained in the component we

called legal and organisational barriers. The results proved that the hypothesis regarding distributors was valid. The most important barriers were classified as organisational, administrative and institutional in the literature review. In Poland, many farmers held negative views about changes to organic farming regulations. This may lead to a greater perceived risk of organic farming compared to conventional farming and discourage the continuation of organic production in the future (Łuczka, 2024). Factors related to the specific nature of organic farming (such as high costs and difficulties in organic production) ranked second. This result is consistent with the studies on barriers identified as economic in the literature review (Grzybowska-Brzezińska & Gorłowa, 2019; Kołoszko-Chomentowska & Stalgiene, 2019). Demand and profitability factors were less significant; however, "low sales opportunities for organic products (insufficient demand)" barrier should not be underestimated. In our opinion, it is due to the small share of organic food in the value of food purchases in Poland - 1% in 2024 (Biokurier, 2025) – which is consistent with the cited research of Michalke et al. (2023). It is also associated with high prices of organic products compared to conventional food (Grzybowska-Brzezińska & Gorłowa, 2019). This factor is more strongly correlated with the fourth component (traditional consumer attitudes) than the other two variables. Distributors place little significance on the common factor describing consumers' preferences, such as "insufficient environmental awareness," as a barrier to sales development.

By combining the results of the exploratory factor analysis and the hierarchy of development barriers based on average measures, it can be concluded that factors related to legal and administrative barriers that determine the functioning of organic farms are the biggest barriers to sales development. This proved that the hypothesis is correct in the light of the opinions of distributors.

In the hierarchy of barriers, based on decreasing average values of measures derived from organic farmers' opinions (Table 4), the most significant factors were bureaucratic and administrative difficulties and the lack of adequate government support. Factors related to the efficiency of organic production (high production costs and low prices for produce offered to organic farms by processors) rank second. The exploratory factor analysis (Table 5) showed that, among the four components, the most significant barriers stemmed from the specific characteristics of organic farming, including high production costs and low profitability due to low prices. This confirmed the results of some studies by other authors (Vlahović et al., 2019; Koreleska, 2017). The low profitability of organic production is also due to price undercutting by supermarket and hypermarket chains (Górska-Warsewicz et al., 2018; Górska-Warsewicz et al., 2021; Łuczka & Kalinowski, 2020). The second most important group of barriers is represented by a common factor relating to legal and organisational barriers, comprising two variables: bureaucratic and administrative difficulties and lack of state support. In this case, the result is similar to that obtained in the survey among distributors.

By combining the results of the exploratory factor analysis and the hierarchy of development barriers based on the average measures, it can be concluded that, taking into consideration farmers' opinions, the hypothesis is partially correct. Factors related to bureaucratic and administrative difficulties and a lack of adequate state support have the highest significance according to the average measure values, but

rank second according to the factor analysis. Institutional and regulatory barriers were crucial for the development of organic farming according to many authors cited in the literature review (Łuczka & Kalinowski, 2020; Komorowska, 2014; Brodzińska, 2014; Nachtman, 2015; Gil, 2016), especially during the period of decline in the number of farms and cultivated area, i.e., between 2013 and 2020 (Naczelną, 2019). As demonstrated, it was due to errors in implementing EU payments for organic farming in Poland.

6 Conclusions

The in-depth analysis of barriers included the exploratory factor analysis allowed the identification of 4 groups of barriers (common factors) according to distributors: legal and organisational ones, factors connected with the specific character of organic farming (difficulties in applying organic farming methods and high production costs), demand and profitability factors, and traditional consumer attitudes. According to the opinions of organic farmers, common factors were partly different: the specific character of organic farming (comprising low yields, difficulties with weed control methods, high costs and low profitability), legal and organisational barriers (the same as in the opinions of distributors, the barriers of organic food processing (little number of processors and low prices of agricultural products), and conventional consumer attitudes. The results of the analyses for both groups of surveyed actors led to the following conclusion: legal and administrative barriers are of primary significance for distributors. According to the hierarchy established using average measures, bureaucratic and administrative difficulties and a lack of adequate state support are the key barriers for farmers. However, according to the factor analysis of the results from that group, the first place is occupied by barriers arising from the specific nature of agricultural production (related to reduced production efficiency). Factors related to inadequate public support rank second but remain very important for farmers. Therefore, the results taken together (for both groups of respondents) proved that the hypothesis was valid. The barriers arising from the high costs and low profitability of organic production are of greatest significance for farmers, given the low prices they receive for agricultural product sales. Therefore, the results taken together (for both groups of respondents) proved that to diminish the barriers, there is a need to simplify and stabilise the system of the European Union's subsidies in Poland. The rules of application for public financial support should be easier and more predictable for farmers. They should have constant conditions for making decisions about organic production and planning its financial results. Taking into account the barriers posed by high costs and low profitability, we concluded that there is a need to increase the rates of organic farming subsidies. It is worth determining in the future to what extent it needs to be raised. The authors intend to undertake research in this direction. It is also worth paying attention to market-led changes, which should involve better connections between farmers and distributors. As our previous research showed, the intensity of farmers'

relationships with food processors is one of the main obstacles to the development of organic food supplies to the market (Kociszewski et al., 2024).

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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